

Reproduction and Extension of DASMCC

Multi-Class Cardiovascular Disease Prediction from 12-Lead ECG

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The paper, and its claim

DASMcC (Sinha et al., *IEEE Access*, 2023) classifies five diagnostic superclasses from PTB-XL 12-lead ECG.

Sixteen time-domain features per record via `neurokit2` - peak positions, peak amplitudes, intervals - then SMOTE and five tree-based classifiers. The pitch is **deployability**: light enough for an IoT device, unlike a deep net on the raw waveform.

21,837

records

5

classes

16

features

93.0 %

reported accuracy

What I set out to do

Follow their method exactly, then explain whatever gap appears.

The reproduction, and the gap

Same features, same classifiers, same metrics. The paper describes *both* a single 80/20 split and 10-fold CV, which cannot both be true, so I implemented both.

Macro-F1	XGBoost	RF	CatBoost	KNN
Published	93.0	92.0	89.0	83.0
Reproduced	39.4	39.2	40.5	25.4

A shortfall of **46 to 58 F1 points**. Class-wise, the paper's easiest class is HYP at 96–99; mine is NORM, and the three minority classes collapse to 21.7–36.0.

A gap that size is normally my own bug. So the next slide is the evidence that it is not.

Three numbers from their own Figure 7

- Their confusion matrix sums to **9,083** test records. A clean 20 % split of the single-label data gives **3,249**.
- Its five row sums are 1,787–1,854 - **near-identical**. Splitting imbalanced data never produces a balanced test set.
- HYP shows 1,796 test records. The entire dataset holds **2,655**.

Conclusion

SMOTE was applied **before** the train/test split. Synthetic minority samples interpolated from training neighbours ended up in the test set.

Reproducing that leakage deliberately lifts RF from 39.2 to 88.3 F1 and recovers 80.5–88.4 % accuracy, against their 88.0–93.0 %. **The gap was never the model.**

What I changed, and why

- **Delineate per lead.** They flatten all twelve leads into one 12,000 vector, then look for P, Q, R, S, T in it. A P wave in V_1 is not a P wave in aVR - the flattening destroys the clinical meaning. I run neurokit2 **on each lead separately**.
- **Add learned features.** Hand-crafted descriptors cannot see what they were not designed for. A 1D CNN reads the raw leads directly: $12 \times 1000 \rightarrow 125$.
- **Then compress.** Concatenating gives $2,100 + 125 = 2,225$ dimensions, where distance-based models suffer. An autoencoder compresses to 16.

The discipline that makes it honest

Both extractors are fitted on the **80 % training split only**, with the same seed the notebooks hold out. That is precisely the mistake I accused the paper of.

Results - against my baseline, not their number

Macro-F1	CatBoost	XGBoost	GBoost	RF	KNN
Reproduction	40.5	39.4	40.5	39.2	25.4
Proposed	58.4	57.5	57.1	56.8	47.9

Every classifier improves: **+16.6 to +18.1** F1 for the ensembles, **+22.5** for KNN - the largest gain, which is what the compression was for. Fitting is also 2.6–9.8× faster.

What did not improve

HYP stays at 21.7–37.3 F1 and *falls* for three of six models. Precision sits far below recall: every model over-predicts the rarest class.

Limits, and what this is worth

- **My 10-fold numbers are not fully clean either.** The extractors were fitted once on 80 %, so CV folds contain records they have seen. I report the held-out split alongside: the gap is **1.0–2.3 points**, which bounds the effect.
- **58 F1 is not a product.** It is roughly two thirds of the way to a usable classifier, and saying so plainly matters more than the number.
- **Single seed, no hyperparameter search.** Differences under a point are not claims I make.

The finding

The published 93 % is an artefact of resampling before splitting, not a property of the method. An honest pipeline on the same data and the same classifiers reaches 58 - and that number is the useful one.